MIL-U-85067A(AS)
17 September 1984
SUPERSEDING
MIL-U-85067(AS)
13 July 1977

MILITARY SPECIFICATION

ULTRASONIC EVALUATION OF HOUSING WELDS

FOR

THE MARK 71 MOD 0 WARHEAD

This specification is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the requirements for detecting discontinuities by ultrasonic inspection of the forward and aft circumferential welds of the Housing (Drawing 2603971) referred to herein as the "housing". This specification also contains the minimum requirements necessary to qualify nondestructive test inspection personnel, procedures, and equipment.

1.2 Classification. The type of ultrasonic evaluation shall be of the pulse-echo technique and immersion method using the compressional (straight beam) mode of acoustic energy.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, handbooks and drawings. Unless otherwise specified, the following specifications, standards, handbooks and drawings of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-P-46838	Potting Compound, Silicone Rubber, Room Temperature Vulcanizing

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Engineering Specifications and Standards Department (Code 93), Naval Air Engineering Center, Lakehurst, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS (Continued)

Naval Air Systems Command
(Code Ident 30003)

AS-1067	Critical Item Product Fabrication Specification Warhead, Guided Missile, MARK 71 MOD 0, Empty

STANDARDS

Military

MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-410	Nondestructive Testing Personnel Qualification and Certification (Eddy Current, Liquid Penetrant, Magnetic Particle, Radiographic and Ultrasonic)
MIL-STD-45662	Calibration System Requirements

DRAWINGS

Naval Air Systems Command
(Code Ident 30003)

2603971	Housing

Naval Weapons Center,
China Lake (Code
Ident 12934)

SK 917AS737	Gage, Housing

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

American Society for Testing and Materials (ASTM)

ASTM E 317	Evaluating Performance Characteristics of Pulse-Echo Ultrasonic Testing Systems

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

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SAE

Aerospace Material Specifications (AMS)

AMS 2632 Ultrasonic Inspection of Thin Material

(Application for copies should be addressed to SAE, 400 Commonwealth Drive, Warrendale, PA 15096.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Equipment. The equipment required for the ultrasonic inspection of the housing shall consist of an ultrasonic test system comprised of subsystems for weld flaw detection and location.

3.1.1 Weld flaw detection subsystem. The weld flaw detection subsystem shall be capable of evaluating forward and aft circumferential weldments as described herein. This subsystem shall include, but not be limited to, the following:

(a) Ultrasonic inspection unit with flaw detection module (see 3.1.4 and 3.1.5).

(b) Ultrasonic search unit(s) (see 3.1.7).

(c) Couplant (see 3.1.8).

(d) Auxiliary equipment (see 3.1.9).

(e) Reference standard (see 3.1.10).

3.1.2 Weld location subsystem. The weld location subsystem shall be capable of locating the position of the forward and aft welds by measuring the distance (L) shown in Figure 1 with an accuracy of ±0.002 inch. This subsystem shall include, but not be limited to, the following:

(a) Ultrasonic inspection unit with thickness measuring module (see 3.1.4 and 3.1.6).

(b) Ultrasonic search unit(s) (see 3.1.7).

(c) Couplant (see 3.1.8).

(d) Auxiliary equipment (see 3.1.9).

(e) Reference standard (see 3.1.10).

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3.1.3 System characteristics. Each ultrasonic test subsystem of the system shall be capable of meeting the Table I requirements as determined by the procedure outlined in ASTM E 317. Where the ultrasonic inspection unit usable Cathode Ray Tube (CRT) screen dimensions exceed 2.0 inches (vertical) and 2.5 inches (horizontal), ASTM E 317 procedures and Table I are still applicable. In this case, the unit shall be considered usable if the entire screen height or width is found to be linear in accordance with ASTM E 317 procedures.

TABLE I. Equipment Characteristics

Instrument Characteristic	Requirements 1/
Vertical limit	2.0 inches (minimum)
Upper linearity limit	2.0 inches (full scale)
Lower linearity limit	0.2 inch (maximum)
Ultrasonic sensitivity	100 percent
Signal-to-noise ratio	100:1 (maximum)
Entry surface resolution 2/	0.100 inch (maximum)
Back surface resolution 2/	0.100 inch (maximum)
Horizontal limit	2.5 inches (minimum)
Horizontal linearity	2.5 inches (full scale)

1/ The requirements shall be as shown and shall be determined for each frequency, pulse repetition rate, and search unit utilized in the ultrasonic test subsystem.

2/ Based on steel.

3.1.4 Ultrasonic inspection unit. The ultrasonic inspection unit (or units) shall be capable of producing, receiving, and amplifying high frequency signals between 5 megahertz (MHz) and 15 MHz and displaying these on a CRT as video signals. The unit(s) shall be of a pulse-reflection (echo), pulse-transmission type. The unit shall have a minimum dynamic range of 30 decibels (dB) and shall have a transmit/receive crosstalk separation of not less than 40 dB. Also, the instrument shall be equipped with a power source voltage regulator which maintains signal amplitude and sweep length within ±5 percent.

3.1.5 Flaw detection modules. Flaw detection modules shall be equipped with vertical and horizontal gates and an audible alarm.

3.1.6 Thickness measuring module. The thickness measuring module shall be capable of measuring the time interval between any two adjacent signals displaying the results as a numerical value with an accuracy of ±0.002 inch. The module shall have a means of calibrating the time (distance) scale of the ultrasonic inspection unit to determine the velocity of the appropriate frequency of acoustic energy in the housing material.

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3.1.7 Ultrasonic search units. The ultrasonic search units (transducers) shall meet the requirements of AMS 2632. Only immersion style search units shall be used. Axial and cross-sectional profiles shall be obtained initially, at 6 month intervals, or at any time search units appear erratic to determine compliance with AMS 2632.

3.1.8 Couplant. An immersion couplant conforming to AMS 2632 shall be used for all ultrasonic evaluation.

3.1.9 Auxiliary equipment.

3.1.9.1 Search unit cables. Coaxial cables and associated fittings shall be physically compatible with the search unit and inspection unit fittings. The coaxial cable should be such that optimum impedance matching is obtained between the inspection unit and the search unit. Impedance matching devices may be used in conjunction with the coaxial cable.

3.1.9.2 Immersion tank. The immersion tank shall be of sufficient size and strength to contain the housing to be inspected, the couplant, and the scanning apparatus. The container shall have filling and drainage capabilities.

3.1.9.3 Scanning apparatus. The materials selected for these apparatus shall be compatible with the couplant and housing.

3.1.9.3.1 Housing support device. The scanning apparatus shall have the capability of accurately supporting the housing with the axis of the housing oriented vertically and in such a manner that the housing can rotate past the search unit. The water path distance from the search unit to the housing on the support device, shall remain constant within $\pm$0.025 inch and the acoustical sound beam shall be perpendicular to the weld surface.

3.1.9.3.2 Search unit fixture (weld evaluation). The search unit fixture for weld evaluation shall be supported by mechanical devices to allow fine adjustment of angle in two planes, perpendicular to the propagation of the longtudinal wave, and to allow accurate positioning of the search unit in the vertical plane. Control and measurement of the search unit angle, water path distance, and vertical position of the search unit, shall be provided by mechanical or electrical means.

3.1.9.3.3 Search unit fixture (weld location). The search unit fixture for weld location shall be supported by mechanical devices to allow fine adjustment of angle in two planes, perpendicular to the propagation of the longitudinal wave, and to allow positioning of the search unit in the horizontal plane. Control and measurement of the search unit angle, water path distance, and horizontal position of the search unit shall be provided by mechanical or electrical means.

3.1.9.3.4 Couplant intrusion fixture. The weldment housing device shall be used with a couplant intrusion fixture to prevent couplant from reaching and entering the rod bundle during inspection of the aft weldment.

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3.1.10 Ultrasonic reference standards. Ultrasonic reference standards shall be used to establish the performance of the inspection system and to interrelate the test results with known reflectors. The ultrasonic reference standards shall consist of production warheads which conform to Drawing 2603971. The forward and aft weld reference standard notches shall be fabricated in accordance with Drawing SK 917AS737.

3.1.10.1 Material. Prior to fabricating the reference standards, the housing to be used for ultrasonic reference standards shall be ultrasonically evaluated at the frequency selected for use. The material shall not contain any weld defects within 1 inch of any reference notch which reduce the axial length of the weld below the minimum acceptable value.

3.1.10.2 Reference reflectors. Reference reflectors (the machined notches) shall be manufactured by Electrical Discharge Machining (EDM) methods. Reference reflectors notches shall be filled with a potting compound conforming to MIL-P-46838, Type IV, and used with the manufacturer's recommended primer to improve adhesion. Degassing of the potting compound shall be performed at 27 inches minimum vacuum prior to use.

3.2 Materials. Not applicable (NA).

3.3 Required procedures and operations.

3.3.1 Written procedure. The contractor shall prepare a written procedure which shall be subject to the approval of the procuring activity for the procedures and operations specified herein. Approval of the written procedures will be by Government personnel capable of preparing, conducting and grading examinations for qualifying Level III ultrasonic inspection personnel in accordance with MIL-STD-410. The procedures shall be sufficiently detailed such that Level II, MIL-STD-410, personnel could duplicate the evaluation and obtain equivalent information. The written procedure(s), as a minimum, shall include the following:

(a) Ultrasonic inspection unit(s) and associated module(s) (manufacturer, model and serial number) (see 3.1.4, 3.1.5, and 3.1.6).

(b) Ultrasonic search unit data (frequency, size, model and serial number) (see 3.1.7).

(c) Couplant data (see 3.1.8).

(d) Auxiliary equipment data (see 3.1.9).

(e) Ultrasonic reference standards used (serial numbers and delta factor) (see 3.1.10).

(f) Detailed standardization and evaluation procedures.

3.3.2 Operations.

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3.3.2.1 System standardization. The weld flaw detection subsystem shall be standardized by establishing a rejection amplitude level as described in 3.3.2.1.1. The test equipment, once standardized, shall not be further adjusted in any way prior to, or during, the actual test, except as specified in 3.3.2.1.2. Standardization shall be required for any of the following conditions:

(a) At the beginning of each shift.

(b) After change of inspection personnel.

(c) After each 10 housings inspected.

(d) After 1 hour of elapsed time.

(e) After loss of power or shutdown for any reason.

(f) Any time inspection personnel have indication of erratic operation of any part of the ultrasonic test system.

3.3.2.1.1 Rejection amplitude level. The reference standard shall be placed in the housing support device for use in standardizing the system. The reference standard shall be rotated such that the acoustic beam is consecutively centered over each of the reference reflector notches. The search units shall be positioned such that reflections from each of the four reference reflectors are clearly distinguished at the baseline of the CRT screen trace; average responses from notches 1 and 1 and average responses from notches 2 and 2 shall be distinguished by minimum amplitude difference of 15 percent CRT screen height. Instrument sensitivity shall be adjusted such that the average signal amplitudes from the four reference reflector notches is 60 to 80 percent of CRT screen height; this amplitude shall include the appropriate delta factor to establish the rejection sensitivity level, which shall be recorded.

3.3.2.1.1.1 Time base and gate adjustment. The horizontal time base and the alarm gate shall be adjusted to display the inspection zone depth. The alarm amplitude shall be adjusted to the rejection amplitude level determined using 3.3.2.1.1.

3.3.2.1.2 Standardization transfer. The location of the aft weld on the housing being evaluated may differ from that of the reference standard. The weld location subsystem shall be utilized to determine the amount of difference between the standard and the housing being evaluated. The distance (L) (see Figure 1) on the standard shall be determined ultrasonically and compared to the distance (L) measured on the housing. The vertical position of the weld flaw detection subsystem search unit shall be changed an appropriate amount and direction. Calibration (see 3.1.6) shall assure proper time scale of the weld location subsystem. The measurement of (L) shall be accomplished by directing the acoustic beam into the standard or housing at the thread seating surface and reflecting from the interface with the rod bundle. Six readings shall be taken at each of the four unthreaded sections and averaged to determine (L).

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3.3.2.2 Scanning speed. The scanning speed shall be regulated so that the reference reflectors are clearly visible on the CRT screen. If distortion related to scanning speed is observed, the scan speed shall be reduced until the distortion is eliminated.

3.3.2.3 Inspection zones. The inspection zones shall include the entire forward and aft circumferential welds beginning at 0.10 inch below the weld surface and extending into the weld root and heat affected zone.

NOTE: The location of the forward and aft welds may vary slightly from the reference standard and shall be compensated for as described in 3.3.2.1.2.

3.3.2.4 Housing evaluation. Housings to be evaluated shall conform to Drawing 2603971. The housing shall be placed in the housing support device of the standardization ultrasonic test system with the axis of the housing vertical. The inspection zones of the forward and aft welds shall be scanned over the entire circumference. Indications from the weld zone, equal to or greater than the rejection amplitude level, shall be noted. The length of each noted defect shall be determined by measuring the circumferential distance, to the nearest 0.1 inch, center-to-center of the acoustic beam along the housing surface corresponding to those points where the amplitude drops below the reject amplitude level. The minimum assigned defect length shall be 1/8 inch. The location and length of all areas equaling or exceeding the criteria shall be recorded (see 6.3). Rejection levels/"pips" less than 0.125 inch shall be recorded for information and identification (see 6.3). Weld quality shall be evaluated in accordance with AS-1067.

3.4 Recommended procedures and operations.

3.4.1 Operations. Automatic or manual systems for ultrasonic evaluation of housing are acceptable for use.

3.5 Certification.

3.5.1 Personnel. Nondestructive testing personnel shall be certified in the ultrasonic method in accordance with MIL-STD-410. Personnel shall be subjected to a practical examination in the operation and standardization (as applicable) of the ultrasonic test system. Recertification of personnel at Level I special and Level II shall be accomplished at 12-month intervals while actively engaged in the ultrasonic inspection process on a regular and continuous basis. Whenever a certified inspector is not directly engaged in the ultrasonic inspection process for a continuous period of 90 days or more, recertification shall be required. Level I special personnel may standardize a system only in the presence of, and under the direct supervision, of a Level II or Level III certified individual. The contractor's personnel certification program shall be submitted for approval by the procuring activity prior to evaluating housings (see 6.3). All certification records shall be readily available for inspection by Government personnel (see 6.2).

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3.5.2 Process. The contractor's process certification program shall be submitted for approval by the procuring activity prior to evaluating housings. The contractor shall submit objective quality evidence of destructive tests which verify the accuracy of the nondestructive ultrasonic evaluation process on housings as required herein (see 6.3).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Monitoring procedures for equipment used in process. Equipment used in the evaluation process shall be calibrated in accordance with MIL-STD-45662, at intervals of 6 months maximum. This calibration shall include the requirements of 3.1.4, 3.1.5, 3.1.6, and 3.1.9. The characteristics of 3.1.3 shall be verified at 6 month intervals which shall coincide with the MIL-STD-45662 calibration and the requirements of 3.1.7. Failure or replacement of any component of a subsystem between regular calibration or verification intervals shall require the affected subsystem to be subjected to the above requirements.

4.3 Monitoring procedures for materials. NA.

4.4 Certification. The contractor shall have available written certification accompanied by objective quality evidence (as defined in MIL-STD-109) that personnel and equipment have been certified in accordance with the requirements specified herein (see 3.5.1 and 3.5.2).

4.4.1 Ultrasonic reference standards. Ultrasonic reference standards fabricated in accordance with the requirements of this specification shall be submitted to the Government for certification. Serial numbers and delta factors (determined by ultrasonic comparison to master ultrasonic reference standards maintained at Weapon Quality Engineering Center, Naval Weapons Station, Concord, CA) shall be used as required by 3.1.10 and 3.3.2.

4.5 Test methods. The contractor shall prepare test procedures to meet the requirements specified in 3.3.1. The procedures shall also be in accordance with the requirements of ASTM E 317 and the applicable requirements of AMS 2632, and shall include a written record describing the performance characteristics of the ultrasonic testing system under a given set of conditions. The test procedures shall be submitted to the procuring activity for approval prior to evaluation of the housings (see 6.3).

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5. PREPARATION FOR DELIVERY.

5.1 Preservation. Housings shall be cleaned and dried of all couplant material upon completion of the ultrasonic evaluation.

6. NOTES

6.1 Intended use. When ultrasonic evaluation of MARK 71 MOD 0 warheads is required, the process outlined in this specification should be used.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Certification records (see 3.5.1).

6.3 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approval CDRL incorporated into the contract. When the provisions of DAR 7-104.9 (n) (2) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs:

Paragraph No.	Data requirement	Applicable DID No.
3.3.1 and 4.5	Test Procedures	DI-T-2072*
3.3.2.4	Inspection Data	DI-T-2072*
3.5.1	Personnel Certification Program	DI-T-2072*
3.5.2	Process Certification Program	DI-T-2072*

*(Paragraph 10.2 does not apply)

(Copies of data item descriptions required by contractors in connection with specified acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

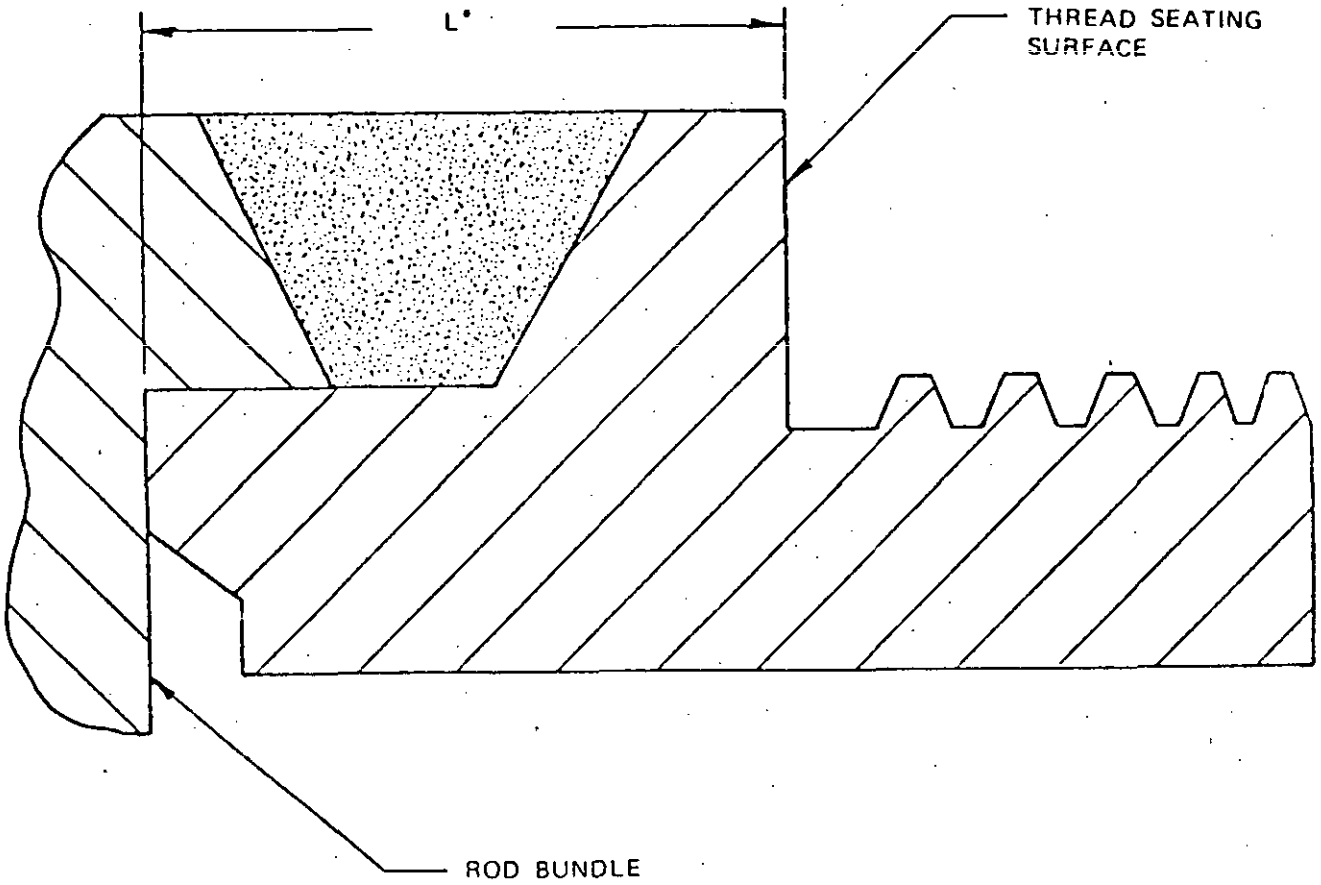
6.4 Supersession data. This specification includes the requirements of Purchase description AS4534 dated 19 May 1976.

6.5 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Preparing activity:
Navy - AS

(Project No. 6635-N121)

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*DIMENSION (L) SHALL BE ULTRASONICALLY DETERMINED FOR EACH WARHEAD TO BE INSPECTED.

FIGURE 1. Sparrow Mark 71 Mod 0 Weldment Showing the Interface Between the Thread Seating Surface and the Rod Bundle.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions - Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-U-85067A(AS)	ULTRASONIC EVALUATION OF HOUSING WELDS FOR THE MARK 71 MOD 0 WARHEAD

3a. NAME OF SUBMITTING ORGANIZATION

b. ADDRESS *(Street, City, State, ZIP Code)*

4. TYPE OF ORGANIZATION *(Mark one)*

- ☐ VENDOR
- ☐ USER
- ☐ MANUFACTURER
- ☐ OTHER *(Specify)*. _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording.

c. Reason/Rationale for Recommendation.

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

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